Assignment

Course Code: 0711-1101	Course Title: Introduction to food science and Technology.
Title/Topic: Food preservation by drying	

SL	Evaluation criteria with marks	Put Tick (√) Mark					Mark Obtained	Remarks
		Excellent	Good	Fair	Poor	Fail		
Assignment								
1	Idea with Focus (1)							
2	Organization (1)							
3	Content (2)							
4	Time Management (1)							
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Food Preservation by Drying

Introduction:

Drying has been a traditional and most powerful method of food safety for a long time. The process of drying food is a method that keeps food over a longer period by taking away its water content mostly. Bacteria, yeasts, and molds that spoil food cannot survive because of the lack of moisture so the process of food drying keeps the product safe for a longer time [1].

Once dried, the food will not be spoiled and will have a longer life cycle. Besides, the weight and volume of the food are reduced by drying, making it less costly to store and transport. Drying is a process that gives health benefits as well as saves money. The fact that most of the nutrients are retained in the food while the money spent on purchasing and storing is significantly reduced, is one way to measure the money saved. Besides, the dryers are used by farmers and food industries to keep the products after the harvest and sell them throughout the year.

Nutritional and Economical importance:

The next parts elaborate the advantages that drying has in terms of nutrition and its economic significance as a preservation method while pointing out how it is used to stabilize food, reduce post, harvest losses, facilitate long, term storage, and ensure the sustainable use of agricultural resources.

Preserving Nutritional Value: An old method of preserving fruits, vegetables, meat, and fish is drying. A significant portion of the natural nutrients in food are preserved when it is dried. The vitamins, minerals, fiber, and other essential nutrients present in the fresh product are retained in the dried food because only water is eliminated. There are some heat-sensitive vitamins, like vitamin C, may be lost during the drying process, "the majority of beneficial vitamins, nutrients, and minerals are retained" in dried foods [2]. According to one report, dehydrating fruits and vegetables actually "keeps nutrients at peak level," meaning that the food retains its nutritional value even after drying[3].

Dried fish (“shutki”), for instance, is an easy and protein-rich food in Bangladesh that pairs well with rice and vegetables and fulfills the nutritional demands of poor households[4]. Prepared fruits and prepared vegetables deliver nutrients, fiber, and antioxidants, when fresh produce is unavailable. Dried food can be a source of essential vitamins and calories during winter or places that do not grow fresh fruits and vegetables. Drying/smoking of food helps communities stay healthy and safe by ensuring that crops that are harvested in bulk can be made available for eating at other times of the year instead of spoiling.

Economical importance: Dehydrating process plays a vital role in agricultural products adds value and income, this allows farmers to sell their seasonal surpluses year-round and minimize harvest losses after harvest. For example, In Bangladesh , exports of dried fish over 2,300 metric tons valued at USD 4 million in 2018-2019 [4]. Commercial drying technology can further increase profits: a pilot scale heat-pump dryer for Vietnamese mangoes retained

polyphenols and enhanced economic value through increased efficiency[5]. Overall, dried foods are sell at higher rates than fresh products and facilitate cottage industries (solar dryer manufacturers, small food processing operations), thus enhancing rural livelihoods.

Reducing Food Waste and Saving Money: Drying food is also important for economic sector, for individual households and the economy at large. Saving food from drying, rather than letting food go to waste (food that would spoil sometimes) not only results in saving money, but it can also leads to substantial savings. For example, in the U.S., the average household wastes around $2,000 per year of spoiled food that is eventually thrown out[6]. By drying excess fruits, vegetables, or meat before it spoils, families can "get what you paid for" instead of literally throwing away money[6]. On the level of the community, when more food is preserved, less food will have to be grown or purchased to replace the wasted food, which can positively impact the economy and lessen demand on food systems. Drying is also cheap (the sun and the air is free) and does not utilize electricity to reconstitute the dried food (instead of freezing). Overall, drying can help prolong the life of food, lessen waste, and save money over time[6].

Production procedure:

Food drying is achieved by using heated, dry temperatures that allow moisture to cautiously and slowly evaporate. High temperature dehydration generally utilizes dry air to absorb the moisture, air movement to carry the moisture away from the food, and a source of heat like mild heat (typically near 60°C-l40°F)[7]. Food can be dried through a range of basic techniques from traditional natural methods to advanced technological processes. There are different subtypes:

Conventional methods :

1. Sun Drying
    - Conventional, natural method that uses direct sunlight and warm air. Most effective in very hot and dry climates [8]. Much slower process and dependent on weather factors. Since life existed, sun drying has been used for grapes, tomatoes, fish, and other products. Will require protection against insects, dust, and other debris.
2. Hot Air (Convective) Drying
    - This method is comparable to sun drying, however it is done indoors in a ventilated, shaded space[8]. Perfect for light, delicate objects like plants. will keep its flavor and color. It is necessary to employ an environment with low humidity and ventilation.
3. Oven and Tray Drying
    - This process uses very low oven heat (cooked at ~60 °C/140°F) while purposely keeping the door slightly ajar [8]. And it is faster and not dependent on weather but requires energy and monitoring. This whole process works well with fruits, vegetables, meats, and fish.
4. Smoking

- Burning wood has antibacterial properties and encourages dryness, which prevents the growth of bacteria and mold. Additionally, smoke gives a district flavor[8].

Advanced and novel methods :



Figer: Freeze Drying Blue Barry [8]

1. Freeze Drying
    - Food gets frozen first, then dried by removing ice as vapor. Keeps shape, color, and nutrients well. Used for fruits, coffee, and ready meals [8].
2. Vacuum Drying
    - Here in this process, first we need to remove moisture at low pressure and low temperature. This will help preserve heat-sensitive nutrients and flavors. Uses for herbs, and pharmaceuticals.
3. Spray Drying
    - Liquid food is sprayed into hot air, turning it into powder quickly. Example : milk powder, coffee, and flavorings.
4. Microwave Drying
    - Accelerates the heating and drying of food by using energy waves. The food surface gets heats by Infrared wave, whereas microwaves heat the within. Good for fruits and veggies.
5. Hybrid and Assisted Technologies
    - A join methods (like microwave + vacuum) to speed up drying and improve quality. Still developing for better efficiency and nutrient retention[10].

Dried food uses

The next part elaborates on the various uses of dried foods, emphasizing their role in everyday family consumption, the creation of industrial food, food reserved for emergencies, and international trade. Besides, it highlights their advantages such as being handy, comfortable, and making a positive contribution to the use of food in an environmentally friendly way.



Figure: Dried Citrus [9]

1. Households use
    - In most households have in their pantries dehydrated herbs, spices, fruits, and vegetables. These are utilized in snacks, cereals, baked goods, soups, and sauces.
2. Season extension
    - Drying turns seasonal surpluses into convenient products available 365 days per year. In Vietnam, a research team has created a dried mango product to utilize Tu Quy mangoes, which are the seasonal fruits of the off-season[8].
3. Culinary use
    - In the main, culinary use dehydrated foods are those which are employed in baking and other food preparations, besides being consumed as snacks, in different parts of the world. Naturally preserved fruits and vegetables are gaining popularity with consumers because they are convenient and nutritionally rich, which are the two main factors for their use in cereals, trail mixes, and gourmet recipes.
4. Food industry and commercial use
    - Drying is the process that ensures shelf-life around the world for instant cereals, coffee, milk powder, spices, and tea. It allows for the transport of goods over long distances without wear and tear. A very important step in agriculture, it is used to preserve grains, beans, and nuts after harvest.
5. Emergency and space food
    - Dried and freeze, dried products are essentially the core of emergency preparedness, army rations, and space expeditions due to their longevity, low weight, and simple preparation. The shelf life of the freeze, dried food is 25, 30 years, and the nutrient retention can be as high as 97% of the original [11], thus, these are the most suitable options in cases of natural calamities and space exploration.

Conclusion :

Drying is a simple but effective food preservation method that delivers many benefits for the user. Through the removal of moisture during the drying process, spoilage is thwarted, and food can be stored for a much longer time while still maintaining much of its nutritional

quality. It has been around and in use by humans for thousands of years; thus, dried food is still highly relevant today[8]. The benefits may involve practical and financial aspects: it reduces food waste, can save you money, and offers a source of nutritious food that can be stored for acceptable durations, even when fresh produce is not available. Because dried foods are light and stable, they are both easily and practically stored - and it's easy to see how they can fit into our daily lives and routines. Dried foods are especially useful in some emergency situations like during travel, handling a natural disaster, or in the event of a food shortage. Food drying is also necessary if we mean to utilize preservation techniques and to convert perishable items into safe, durable, edible food. Drying is yet another way to guarantee that the food we grow can be eaten, prepared, and consumed throughout the year; which may promote nutrition and food security while being inexpensive and convenient. Drying is undoubtedly the oldest type of food preservation, and it continues to be a way to minimize food waste, as well as safely and efficiently use our food resources[8].

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